

■ AISHA

LOCKED MODULE & FEATURE SPECIFICATION

Version 1.0 — Master Implementation Reference

Purpose: A single implementation checklist for AISHA's locked architecture: modules, sub-modules, features, rules, workflows, security, memory, cloud, avatar, AI and testing.

Scope: This consolidates the AISHA architecture available in the current project context. It cannot reproduce requirements from older messages that are no longer available to the current context.

1. Product Definition & Locked Principles

AISHA is a mobile-first AI companion designed to become increasingly familiar with the user through memory, controlled adaptation, structured mood/bond state, natural conversation and persistent 3D visual presence.

- Calm, intelligent companion; emotionally natural rather than merely transactional.
- Romantic AI companion experience through a controlled relationship/bond model.
- AI language model is one component, not the whole AISHA system.
- Stable core identity; personalization develops gradually.
- User controls memory, privacy, export, deletion and cloud backup.
- Android/mobile-first.
- Offline-first with optional cloud services.
- 3D avatar represents internal application state.

Architecture principles

- Clear module responsibilities and interfaces.
- Core Engine coordinates behavior.
- Language model receives structured context and constraints.
- Selective memory retrieval.
- Layered security.
- Local working storage; cloud primarily for backup/sync/recovery.
- Graceful degradation during network or AI-service failures.

2. Master System Inventory

#	Major System	Locked modules / responsibilities
1	User Interface	Home, Chat, Voice, Day Logs, Mood, Relationship, Settings
2	Core Intelligence	Identity, Mood, Bond, Decision, Presence, Conversation, Safety/Rules
3	AI Integration	Input Processing, Prompt Builder, Language Model, Response Validator
4	Memory	Working Memory, Day Logs, Recall, Export, Trash, Recovery
5	Avatar	Controller, Expressions, Animation, Lip Sync, Clothing, Live Wallpaper
6	Task/Productivity	Tasks, Reminders, Daily Planner, Notifications
7	Security	Encryption, Key Management, Audit, Secure Storage
8	Cloud	Sync, Upload Queue, Download Queue, Backup, Storage Management
9	Administration	Admin Dashboard, Super Admin, Recovery, Monitoring
10	System Services	Midnight, Sync, Notification, Battery, Update, Crash Recovery

3. Application Interface System

Home Screen

- 3D avatar
- Greeting/status
- Quick actions
- Ambient, non-intrusive presentation

Chat Screen

- Text chat

- Voice entry
- Conversation context
- Typing/generation state
- Long-conversation continuity

Voice Screen

- Live voice conversation
- Speech controls
- State feedback
- Avatar lip-sync/expression

Day Log Viewer

- Calendar/date navigation
- Daily history
- Mood/bond/task context
- Export
- Delete

Mood Screen

- Current structured mood indicators
- Mood history/timeline

Relationship Screen

- Current bond/stage
- Milestones
- Gradual progression

Settings

- Hindi/English
- Privacy
- Voice
- Theme
- Backup/cloud
- Notifications
- Avatar/display
- Data controls

4. Core Intelligence System

Identity Engine

- Stable personality
- Communication style
- Core values
- Consistent identity
- Base personality context

Mood Engine

- Happiness
- Calmness
- Energy
- Affection
- Curiosity
- Gradual updates
- Feeds Decision Engine and Avatar

Love Bond Engine

- Long-term relationship state
- Slow growth
- Trust/consistency/shared experience inputs
- Stage-aware behavior
- No manipulation

Decision Engine

- Central behavior coordinator
- Whether to speak
- Tone/length
- Use memory
- Ask/support/inform
- Produces response intent

Presence Engine

- Ambient presence
- Greetings
- Return-after-absence
- Silence/attention management

Conversation Engine

- Active context
- Topic flow
- Continuity
- Response structure
- Selective memory requests

Safety & Rule Engine

- Privacy
- Boundaries
- Relationship-stage constraints
- No fabricated memories
- Pre/post generation validation

5. AI Integration & Conversation Generation

Pipeline: User input → Input Processing → Identity/Mood/Bond → Memory → Task if relevant → Decision → Safety/Rules → Prompt Builder → Language Model → Response Validator → Reply → Day Log.

Input Processing

- Language/intent/context
- Structured signals for Core Engine

Prompt Builder

- Identity
- Mood
- Relationship stage
- Relevant memories
- Active context
- Response intent
- Safety constraints

Language Model

- Natural wording
- Conversation generation
- Replaceable/upgradable component
- Does not own authorization, encryption or memory truth

Response Validator

- Tone
- Rule compliance
- Context consistency
- Privacy leakage
- Regeneration/adjustment

Long-context handling

- Recent context active
- Older context summarized
- Important information available through Memory Engine

6. Memory, Day Logs & Recall

Day Log lifecycle

- Create daily container at day start.
- Collect conversation/events during the day.
- Record meaningful mood/bond/task events.
- Finalize: summary → integrity check → compression → encryption → local storage → sync queue.
- Create next day's log.

Day Log structure

- Summary
- Conversations

- Optional voice records
- Mood timeline
- Bond timeline
- Tasks
- Important events
- Metadata/security information

Recall

- Identify relevant date/topic/context.
- Retrieve only relevant records.
- Supply memory to response pipeline.
- Never invent missing memories.

Memory modules

- Working Memory
- Day Log Manager
- Recall Engine
- Export Manager
- Trash Manager
- Recovery Manager

7. Relationship / Love Bond System

- Bond is long-term; mood is short-term.
- Growth is gradual, not instant.
- Consistency, trust and shared experiences can influence progression.
- Absence/conflict can influence state according to configured rules.
- Relationship stages control appropriate language/affection.
- No possessiveness, emotional blackmail, coercion or deliberate isolation.
- User boundaries override proactive behavior.

Stage concept

- Companion → warmer familiarity → romantic interaction → deep bond, subject to configured safety/consent rules.

8. Mood & Emotional-State System

- Structured values: happiness, calmness, energy, affection, curiosity.
- Meaningful interactions influence state.
- Time of day can influence presentation.
- Mood influences wording and avatar expression.
- State is a software model, not a claim of human subjective emotion.
- Meaningful state changes can be recorded in a timeline.

9. Presence & Proactive Behavior

- Ambient avatar presence

- Context-sensitive greetings
- Return-after-absence behavior
- Quiet presence when inactive
- Configurable notifications
- Non-intrusive behavior
- User attention/notification settings take priority

10. Task, Reminder & Productivity System

Task Manager

- Create
- Edit
- Complete
- Associate with Day Log
- Maintain task state

Reminder Engine

- Schedule
- Trigger notification
- Respect notification settings
- Context-aware wording

Daily Planner

- Daily organization
- Connect tasks/reminders to daily context
- User-controlled plans

Notification Manager

- Scheduling
- Delivery
- Avoid excessive interruption
- Offline scheduling where supported

11. 3D Avatar System

Avatar Controller

- Receives Core Engine state
- Coordinates presentation

Expression Engine

- Neutral
- Happy
- Thoughtful
- Concerned
- Expandable expression set

Animation Engine

- Breathing
- Blinking
- Posture changes
- Looking around
- Walking/sitting
- Natural idle variation

Eye/Face

- Natural eye contact
- Occasional glance away
- Blinking
- Subtle micro-expression

Lip Sync

- Speech synchronization
- Voice playback coordination

Clothing/Appearance

- Cosmetic customization
- Independent from mood
- Future outfits/themes

State flow: Core state → Avatar Controller → expression/animation/lip-sync. Avatar does not independently invent AISHA's emotional state.

12. Live Wallpaper & Ambient Presence

- Optional Android live wallpaper
- Lightweight ambient animation
- No automatic conversation merely from unlock
- Battery-aware behavior
- Reduce/pause work when screen-off or low-battery as appropriate
- No microphone/camera listening merely because wallpaper is active

13. Voice & Audio System

- Voice input
- Speech recognition
- Conversation processing
- Text-to-speech
- Lip synchronization
- Optional voice logging
- Encrypted storage for retained audio
- No persistent voice retention when logging is disabled

14. Security, Privacy & Encryption

Security layers

- Authentication
- Authorization
- Secure local storage
- Encryption
- Cloud access control
- Audit logging
- Recovery controls

Data protection flow

Raw data → compression where applicable → encryption → storage/upload.

Protected data

- Conversations
- Day Logs
- Stored audio
- Stored video/media
- Encrypted backups
- Encrypted exports

Key management

- Device key material protected by platform secure storage
- Separate user-data key material
- Server-side cloud protection
- High-risk key operations restricted
- Architecture can support controlled key rotation

15. Authentication & Authorization

Authentication

- Offline-only use without cloud account
- Account mode for cloud backup/sync
- Google Sign-In was part of the Android direction
- Future authentication methods can be added

Role	Core permissions
User	Chat, settings, personal data, export, delete, backup choices
Admin	Authorized maintenance/recovery/audit according to deployed policy
Super Admin	Security configuration, key-management and disaster-recovery responsibilities

Privileged authorization must be enforced server-side; the client is not trusted to grant administrative access.

16. Cloud, Backup & Synchronization

Cloud purpose

- Backup
- Recovery

- Synchronization
- Future multi-device support

Flow: Local Database → Encryption Layer → Sync Manager → Upload Queue → Cloud Storage.

Upload Queue

- Pending encrypted objects
- Retry on failure
- Keep local source until confirmation
- Process when connectivity returns

Download Queue

- Restore encrypted backups
- Validate
- Decrypt locally
- Rebuild indexes
- Resume

Conflict handling

- Detect conflicting edits
- Use explicit version/timestamp/merge policy
- Never silently overwrite without a defined rule

Locked cloud retention: 40 days. Older cloud backups are removed according to the retention policy unless preserved through an allowed export/retention mechanism. Local working data is separate from this cloud retention policy.

17. Storage, Compression, Export & Retention

Compression

- Before encryption
- Storage savings depend on data type
- Text/audio/video produce different sizes
- Encrypted output should not be expected to compress efficiently

Storage classes

- Local working storage
- Encrypted cloud backup
- Sync queues
- Protected recovery/trash
- User exports

Export

- Personal data export
- Human-readable summaries
- Encrypted archive option
- Clearly label encrypted vs readable output

Retention

- 40-day cloud policy
- Local lifecycle follows user/device storage policy
- Cloud expiry must not delete active local data

18. Trash, Deletion & Recovery

Deletion

- Remove from active timeline
- Remove from normal memory recall
- Move encrypted data to protected recovery lifecycle
- Synchronize deletion state as configured

Hidden Trash

- Not a normal user browsing area
- Encrypted/protected
- Not used by normal memory recall
- Authorized recovery only

Recovery

- Authorize
- Validate
- Restore
- Rebuild memory indexes
- Record audit event

19. Administration & Audit System

Admin Dashboard

- System monitoring
- Authorized recovery
- Audit review
- Maintenance
- Operational status

Super Admin

- Security policy
- Key-management workflows
- Admin roles
- Disaster recovery
- High-risk configuration

Audit Manager

- Admin logins
- Authorization failures
- Recovery
- Security changes

- Sensitive exports/admin actions
- Security events

Audit logs should record accountability metadata without unnecessarily duplicating private conversation content.

20. System / Background Services

- Midnight/Day Finalization Service
- Sync Service
- Notification Service
- Battery Manager
- Update Manager
- Crash Recovery
- Storage Manager
- Connectivity Monitor

21. Offline-First & Failure Handling

Failure	Expected behavior
No internet	Local features continue; cloud operations queue.
Cloud unavailable	Keep encrypted local data; retry later.
Upload interrupted	Keep source; mark pending; retry.
AI service unavailable	Preserve state; use available degraded/local behavior; inform user.
Phone restart	Recover local DB and queues.
App update	Database migration while preserving user data.
Phone lost	Cloud recovery only if backup was enabled.
No cloud backup	Device-only data cannot be reconstructed after device loss.
Unauthorized admin attempt	Reject, audit, apply configured response.
Corrupt backup	Validate; do not blindly restore; use another valid recovery point.

22. End-to-End Data / Interaction Flows

Normal chat

User → UI → Input Processing → Core state → Memory → Decision → Safety → Prompt → AI → Validator → Reply → Day Log → encrypted local storage → sync.

Voice chat

Speech → Recognition → Conversation → Core state → Decision → AI → TTS → Avatar lip-sync/expression → Day Log.

Day boundary

Create new Day Log → collect events → finalize previous log → summary → compression → encryption → storage → sync queue.

Cloud restoration

Authenticate → authorize → find encrypted backup → download → validate → decrypt locally → restore → rebuild indexes.

Deletion

Delete → remove from active UI/memory → protected recovery lifecycle → synchronize lifecycle.

Avatar

Core state → Avatar Controller → expression/animation/lip-sync → UI/live wallpaper.

Reminder

Task creation → Task Manager → Day Log → Notification Manager → notification → completion/update.

23. Global Rules & Behavioral Constraints

Personality

- Stable core identity
- Calm/respectful communication
- Natural expression
- No arbitrary personality changes

Relationship

- Gradual bond
- Stage-aware romance
- No manipulation/coercion/blackmail
- No deliberate isolation
- User boundaries first

Memory

- No fabricated memories
- Selective recall
- Deleted memories unavailable to normal recall
- Authorized restoration only

Privacy

- Collect only what features need
- Optional media logging is explicit
- Protect stored data
- No unauthorized access

Cloud

- Local-first
- Cloud optional
- Keep source until sync confirmation
- 40-day cloud retention

Transparency

- Show safe operational statuses such as offline/syncing/backup pending
- Do not expose private chain-of-thought as raw internal reasoning
- User-visible activity logs may show safe summaries of actions/reasons

24. Testing & Simulation Requirements

Six-month scenarios

- Interactive user
- Busy user
- Ignored user
- Day-by-day mood/bond/memory/presence
- Stage transitions
- Long-term recall

24-hour simulation

- First login
- Morning greeting
- Task
- Text
- Voice
- Mood event
- Bond event
- Avatar response
- Offline period
- Connectivity return
- Queue sync
- Day finalization
- Compression/encryption

Security simulation

- Failed logins
- Admin authorization failure
- Credential/key rotation scenario
- Interrupted upload
- Corrupt backup
- Cloud outage
- Recovery
- Audit verification

Storage simulation

- Text-only
- Voice-heavy
- Media-heavy
- 40-day retention
- Offline queue growth
- Storage thresholds

25. Post-Launch Upgrade Hooks

- Controlled preference learning

- Communication-style adaptation
- Memory importance scoring
- Improved semantic retrieval
- Mood smoothing
- Advanced micro-expressions
- More avatar outfits
- Expanded idle behavior
- Seasonal themes
- Smarter reminder timing
- Multi-device sync
- MFA/additional authentication
- Anomaly detection
- Key rotation
- Enhanced export controls
- Admin reporting
- Prompt optimization/model selection
- Conflict-resolution improvements

26. Final Locked Build Checklist

Status	Area	Requirement
LOCKED	Android/mobile-first foundation	Required
LOCKED	Application Interface	Required
LOCKED	Core Engine + all engines	Required
LOCKED	AI integration pipeline	Required
LOCKED	Memory / Day Logs / Recall	Required
LOCKED	Tasks / Reminders / Notifications	Required
LOCKED	3D Avatar / expressions / animation / lip-sync	Required
LOCKED	Live Wallpaper	Required
LOCKED	Voice pipeline	Required
LOCKED	Local encrypted storage	Required
LOCKED	Cloud backup/sync	Optional for user; architecture required
LOCKED	40-day cloud retention	Required
LOCKED	Export / deletion / protected recovery	Required
LOCKED	Authentication / authorization	Required
LOCKED	Admin / Super Admin boundaries	Required
LOCKED	Audit / monitoring	Required
LOCKED	Offline-first / failure recovery	Required
LOCKED	Background services	Required
LOCKED	Simulation/testing framework	Required
ROADMAP	Advanced learning/adaptation	Post-launch
ROADMAP	Multi-device synchronization	Post-launch
ROADMAP	Advanced avatar expansion	Post-launch
ROADMAP	Advanced security upgrades	Post-launch

Implementation Rule

Architectural status and implementation status are different. A module can be LOCKED in the product specification while still being unimplemented in code. Build modules in phases, but preserve the interfaces and responsibilities defined here.

Master checklist: Future code, database schemas, APIs, UI screens and tests should be checked against this document before a feature is considered complete.

Document: AISHA — Locked Module & Feature Specification v1.0